HUNNY TUTORIALS

Your CBSE Partner

Class X Maths Assignment-1

Ch- Area Related to Circles NCERT Exemplar

Multiple Choice Questions

Q.1. If the area of a circle is 154 cm^2, then its perimeter is

(a) 11 cm (b) 22 cm (c) 44 cm (d) 55 cm

Q.2. If θ is the angle (in degrees) of a sector of a circle of radius r, then area of the sector is

(a) $\frac{\pi r^2\theta}{360}$ (b) $\frac{\pi r^2\theta}{180}$ (c) $\frac{2\pi r\theta}{360}$ (d) $\frac{2\pi r\theta}{180}$

Q.3. If the sum of the areas of two circles with radii R_1 and R_2 is equal to the area of a circle of radius R, then

(a) $R_1 + R_2 = R$ (b) $R_1^2 + R_2^2 = R^2$ (c) $R_1 + R_2 < R$ (d) $R_1^2 + R_2^2 < R^2$

Q.4. If the sum of the circumference of two circles with radii R_1 and R_2 is equal to the circumference of a circle of radius R, then

(a) $R_1 + R_2 = R$ (b) $R_1 + R_2 > R$ (c) $R_1 + R_2 < R$ (d) Nothing definite can be said about the relation among R_1 , R_2 , R.

Q.5. If the circumference of a circle and the perimeter of a square are equal, then

(a) Area of the circle = Area of the square

(b) Area of the circle > Area of the square

(c) Area of the circle < Area of the square

(d) Nothing definite can be said about the relation between the areas of the circle and square.

Q.6. Area of the largest triangle that can be inscribed in a semi-circle of radius *r* units, in square units is:

(a) r^2 (b) $1/2r^2$ (c) $2\ r^2$ (d) $\sqrt{2}r^2$

Q.7. If the perimeter of a circle is equal to that of a square, then the ratio of their areas is:

(a) 22:7 (b) 14:11 (c) 7:22 (d) 11:14

Q.8. It is proposed to build a single circular park equal in area to the sum of areas of two circular parks of diameters 16 m and 12 m in a locality. The radius of the new park would be

(a) 10 m (b) 15 m (c) 20 m (d) 24 m

Q.9. The area of the circle that can be inscribed in a square of side 6 cm is

(a) 36 π cm^2 (b) 18 π cm^2 (c) 12 π cm^2 (d) 9 π cm^2

Q.10. The area of the square that can be inscribed in a circle of radius 8 cm is

(a) 256 cm^2 (b) 128 cm^2 (c) 642 cm^2 (d) 64 cm^2

Q.11. The radius of a circle whose circumference is equal to the sum of the circumferences of the two circles of diameters 36 cm and 20 cm is

(a) 56 cm (b) 42 cm (c) 28 cm (d) 16 cm

Q.12. The diameter of a circle whose area is equal to the sum of the areas of the two circles of radii 24 cm and 7 cm respectively, is

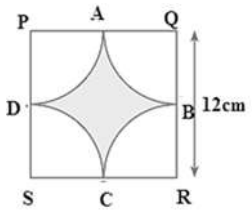
(a) 31 cm (b) 25 cm (c) 62 cm (d) 50 cm

Q.13. If the length of an arc of a circle of radius r is equal to that of an arc of a circle of radius $2r$, then
(a) the angle of the corresponding sector of the first circle is double the angle of the corresponding sector of the other circle.
(b) the angle of the corresponding sector of the first circle is equal the angle of the corresponding sector of the other circle.
(c) the angle of the corresponding sector of the first circle is half the angle of the corresponding sector of the other circle.
(d) the angle of the corresponding sector of the first circle is 4 times the angle of the corresponding sector of the other circle.

Q.14. The wheel of a motor cycle is of radius 35 cm. How many revolutions per minute must the wheel make so as to keep a speed of 66 km/h?
(a) 300 (b) 400 (c) 450 (d) 500

Q.15. A cow is tied with a rope of length 14 m at the corner of a rectangular field of dimensions 20m × 16m, then the area of the field in which the cow can graze is:
(a) 154 m^2 (b) 156 m^2 (c) 158 m^2 (d) 160 m^2

Q.16. The area of the shaded region in Fig., where arcs drawn with centres P, Q, Rand S intersect in pairs at mid-points A, B, C and D of the sides PQ, QR, RS and SP, respectively of a square PQRS, is:



(a) 25.25 cm^2 (b) 27.45 cm^2 (c) 29.65 cm^2 (d) 30.96 cm^2

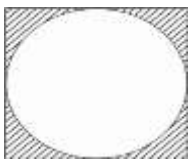
Q.17. Area of a sector of central angle 120° of a circle is 3π cm^2. Then the length of the corresponding arc of this sector is:
(a) 5.8cm (b) 6.1cm (c) 6.3cm (d) 6.8cm

Short and Long Questions

Q.1. Find the diameter of the circle whose area is equal to the sum of the areas of the two circles of diameters 20cm and 48cm.

Q.2. Find the area of a sector of circle of radius 21cm and central angle 120°.

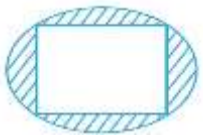
Q.3. In Fig., a circle of radius 7.5 cm is inscribed in a square. Find the area of the shaded region (Use $\pi = 3.14$)



Q.4. Area of a sector of a circle of radius 36cm is 54π cm^2. Find the length of the corresponding arc of the sector.

Q.5. Find the radius of a circle whose circumference is equal to the sum of the circumferences of two circles of radii 15cm and 18cm.

Q.6. In Fig., square of diagonal 8cm is inscribed in a circle. Find the area of the shaded region.

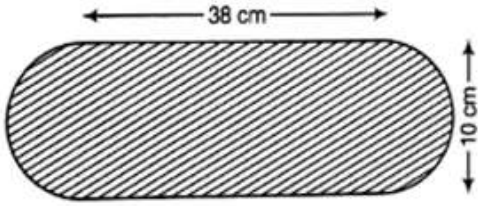


Q.7. Find the area of a sector of a circle of radius 28cm and central angle 45°.

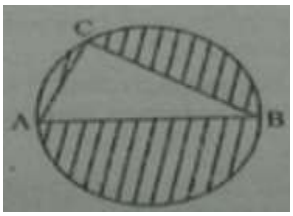
Q.8. The wheel of a motor cycle is of radius 35cm. How many revolutions per minute must be wheel make so as to keep a speed of 66 km/h?

Q.9. A cow is tied with a rope of length 14m at the corner of a rectangular field of dimensions 20m x 16m. Find the area of the field in which the cow can graze.

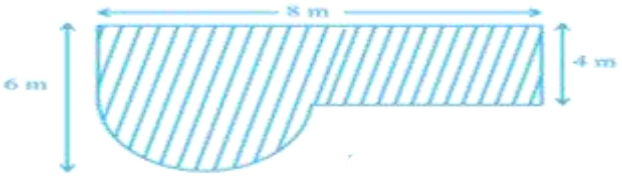
Q.10. Find the area of the flower bed (with semi-circular ends) shown in Fig.



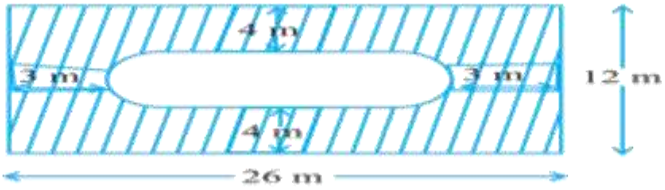
Q.11. In Fig. 4, AB is a diameter of the circle, AC = 6cm and BC = 8 cm. Find the area of the shaded region (Use π = 3.14).



Q.12. Find the area of the shaded field shown in Fig.

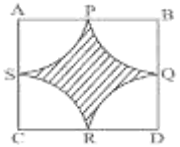


Q.13. Find the area of the shaded region in Fig.

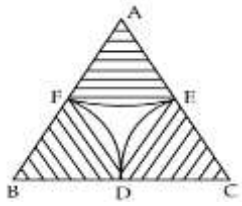


Q.14. Find the area of the minor segment of a circle of radius 14cm, when the angel of the corresponding sector is 60°.

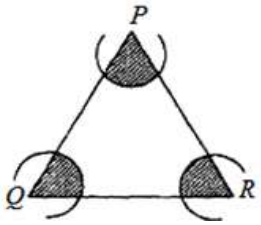
Q.15. Find the area of the shaped region in Fig., where arcs drawn with centres A, B, C and D intersect in pairs at mid-points P, Q, R and S of the sides AB, BC, CD and DA, respectively of a square ABCD (Use π = 3.14).



Q.16. In Fig., arcs are drawn by taking vertices A, B and C of an equilateral triangle of side 10cm. to intersect the sides BC, CA and AB at their respective mid-points D, E and F. Find the area of the shaded regions (Use π = 3.14).

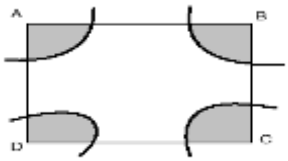


Q.17. In Fig., arcs have been drawn with radii 14cm each and with centres P, Q and R. Find the area of the shaded region.



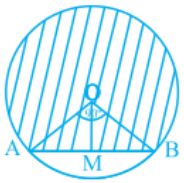
Q.18. A circular park is surrounded by a road 21 m wide. If the radius of the park is 105 m, find the area of the road.

Q.19. In Fig., arcs have been drawn of radius 21cm each with vertices A, B, C and D of quadrilateral ABCD as centres. Find the area of the shaded region.



Q.20. A piece of wire 20cm long is bent into the form of an arc of a circle subtending an angle of 60° at its centre. Find the radius of the circle.

Q.21. A chord of a circle of radius 20 cm subtends an angle of 90° at the centre. Find the area of the corresponding major segment of the circle. (Use π = 3.14).



Q.22. With the vertices A, B and C of a triangle ABC as centres, arcs are drawn with radii 5cm each as shown in Fig., If AB = 14cm, BC = 48cm and CA = 50cm, then find the area of the shaded region. (Use π = 3.14).



Q.23. A calf is tied with a rope of length 6m at the corner of a square grassy lawn of side 20m. If the length of the rope is increased by 5.5m, find the increase in area of the grassy lawn in which the calf can graze.

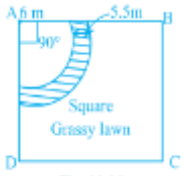


Fig. 11.16

Q.24. The area of a circular playground is 22176 m². Find the cost of fencing this ground at the rate of Rs. 50 per metre.

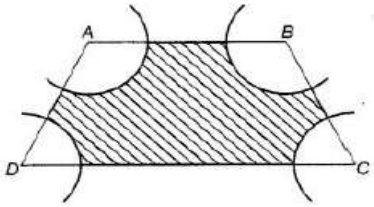
Q.25. The diameters of front and rear wheels of a tractor are 80cm and 2m respectively. Find the number of revolutions that rear wheel will make is covering a distance in which the front wheel makes 1400 revolutions.

Q.26. Sides of a triangular field are 15m, 16m and 17m. With the three corners of the field a cow, a buffalo and a horse are tied separately with ropes of length 7m each to graze in the field. Find the area of the field which cannot be grazed by the three animals.

Q.27. Find the area of the segment of a circle of radius 12 cm whose corresponding sector has a central angle of 60° (Use $\pi = 3.14$).

Q.28. A circular pond is 17.5 m is of diameter. It is surrounded by a 2m wide path. Find the cost of constructing the path at the rate of Rs. 25 per m².

Q.29. In Fig., ABCD is a trapezium with AB ‖ DC, AB = 18 cm, DC = 32 cm and distance between AB and DC = 14cm. If arcs of equal radii 7cm with centres A, B, C and D have been drawn, then find the area of the shaded region of the figure.



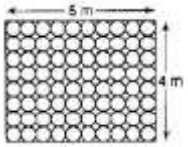
Q.30. Three circles each of radius 3.5cm are drawn in such a way that each of them touches the other two. Find the area enclosed between these circles.

Q.31. Find the area of the sector of a circle of radius 5cm, if the corresponding arc length is 3.5cm.

Q.32. Four circular cardboard pieces of radii 7cm are placed on a paper in such a way that each piece touches other two pieces. Find the area of the portion enclosed between these pieces.

Q.33. On a square cardboard sheet of area 784 cm², four congruent circular plates of maximum size are placed such that each circular plate touches the other two plates and each side of the square sheet is tangent to two circular plates. Find the area of the square sheet not covered by the circular plates.

Q.34. Floor of a room is of dimensions 5m x 4m and it is covered with circular tiles of diameters 50cm each as shown in Fig., Find the area of floor that remains uncovered with tiles. (Use $\pi = 3.14$).



Q.35. All the vertices of a rhombus lie on a circle. Find the area of the rhombus, if area of the circle is 1256 cm^2. (Use $\pi = 3.14$).

Q.36. An archery target has three regions formed by three concentric circles as shown in Fig., If the diameters of the concentric circles are in the ratio 1:2:3, then find the ratio of the areas of three regions.

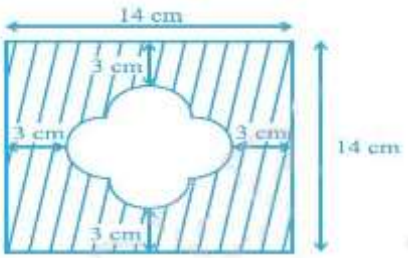


Q.37. The length of the minute hand of a clock is 5cm. Find the area swept by the minute hand during the time period 6:05 a m and 6: 40 a m.

Q.38. Area of a sector of central angle 200° of a circle is 770 cm^2. Find the length of the corresponding arc of this sector.

Q.39. The Central Angles of the two sectors of circles of radii 7 cm and 21 cm are respectively 120° and 40°. Find the ares of the two sectors as well as the lengths of the corresponding areas. What do you Observe?

Q.40. Find the area of the shaded region given in Fig.



Q.41. Find the number of revolutions made by a circular wheel of area 1.54m^2 in rolling a distance of 176m.

Q.42. Find the difference of the areas of two segments of a circle formed by a chord of length 5cm subtending an angle of 90° at the centre.

Q.43. Find the difference of the areas of a sector of angle 120° and its corresponding major sector of a circle of radius 21 cm.

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Class X Mathematics
Practice Test
Ch- Area related to Circles

*Followings multiple choice questions are of **1 Mark** each (Q01-10).*
Select the correct option in each one of them.

Q01. The ratio of the areas of two circles is 9 : 16. The ratio of their circumferences will be
(A) 3 : 4 (B) 9 : 16 (C) 81 : 256 (D) 4 : 3

Q02. The area of a sector of a circle of radius r is $\frac{\pi r^2}{4}$. The angle of the sector is
(A) 60° (B) 90° (C) 120° (D) 180°

Q03. If the perimeter of a semicircular protractor is 72 cm, then its diameter is
(A) 18 cm (B) 24 cm (C) 28 cm (D) 48 cm
[Use $\pi = \frac{22}{7}$]

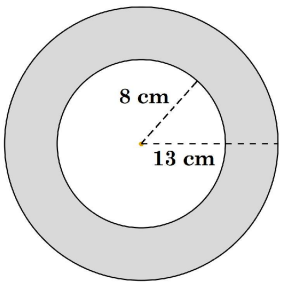
Q04. The difference between the circumference and the diameter of a circle is 30 cm. The radius of the circle is
(A) 14 cm (B) 7 cm (C) 21 cm (D) 28 cm

Q05. A horse is tied to a peg at one corner of a square-shaped grass field of side 15 m by a rope 5 m long. The area of the field in which the horse can graze is
(A) $\frac{25\pi}{4}$ m^2 (B) 75π m^2 (C) 100π m^2 (D) 225π m^2

Q06. A chord of a circle of radius 10 cm subtends a right angle at the center. The area (in cm^2) of the minor segment is
(A) $25\pi - 50$ (B) $50\pi - 100$ (C) $75\pi - 50$ (D) $100\pi - 100$

Q07. The wheel of a car makes 1000 revolutions. If the diameter of the wheel is 84 cm, the distance covered is
(A) 2.64 km (B) 2.44 km (C) 2.64 m (D) 26.4 km

Q08. The radii of two concentric circles are 13 cm and 8 cm as shown below.
The area of the ring (shaded region) between them is
(A) 69π cm^2 (B) 105π cm^2
(C) 169π cm^2 (D) 225π cm^2



*Followings are **Assertion-Reason based questions** (Q09 & 10).*
In the following questions, a statement of Assertion (**A**) is followed by a statement of Reason (**R**).
Choose the correct answer out of the following choices.
(A) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(B) Both **A** and **R** are true and **R** is not the correct explanation of **A**.
(C) **A** is true but **R** is false.
(D) **A** is false but **R** is true.

Q09. **Assertion (A) :** The diameter of a wheel is 1.4 m. To cover 1.1 km, the wheel shall make exactly 250 revolutions, if $\pi = \frac{22}{7}$.

Reason (R) : Circumference of a circle $= \pi\left(\frac{d}{2}\right)$, where d is the diameter.

Q10. **Assertion (A) :** A circular park has a path running around it of width 7 m. If the outer radius of the park is 21 m, the area of the path is $770\ m^2$.

Reason (R) : The area of a segment of a circle is obtained by subtracting the area of the corresponding triangle from the area of the sector i.e., Area of segment = Area of sector – Area of triangle.

$[1 \times 10 = 10$

SECTION B

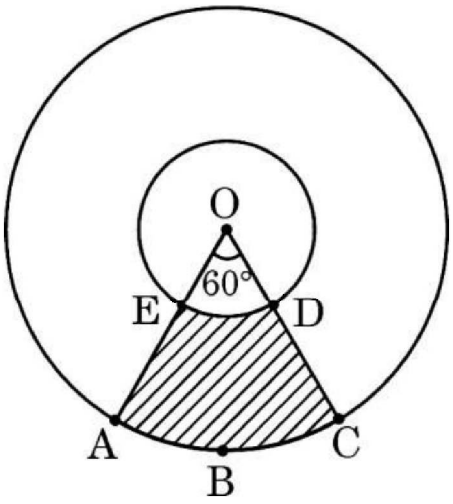
*Followings are of **2 Marks** each (Q11-12).*

Q11. In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find the area of the sector formed by the arc. Also, find the length of the arc.

Q12. (a) A horse is tied to a peg at one corner of a square shaped grass field of side 15 m by means of a 5 m long rope. Find the area of that part of the field in which the horse can graze. Also, find the increase in grazing area if length of rope is increased to 10 m. (Use $\pi = 3.14$).

OR

(b) In the given figure, two concentric circles with centre O are shown.
Radii of the circles are 2 cm and 5 cm respectively.
Find the area of the shaded region.



$[2 \times 2 = 4$

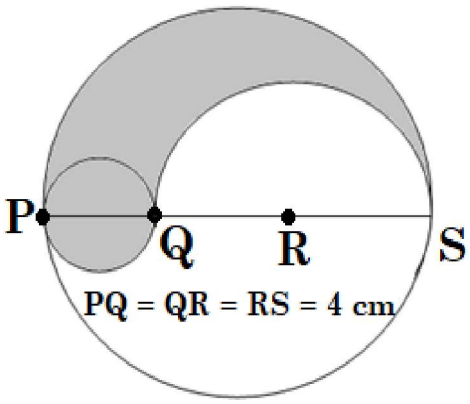
SECTION C

*Followings are of **3 Marks** each (Q13-16).*

Q13. (a) Find the area of the sector of a circle of radius 7 cm and of central angle 90°. Also, find the area of corresponding major sector.

OR

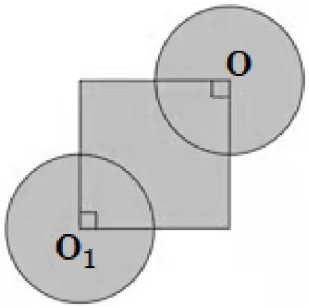
(b) In the given figure, find the area of shaded region.



Q14. A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. Find the area of the corresponding minor segment of the circle. Also find the area of the major segment of the circle.

Q15. Two circles touch each other externally and the sum of their areas is $52\pi\ cm^2$. If the distance between the centers of two circles is 10 cm, find their radii.

Q16. In the given figure, the side of square is 28 cm and radius of each circle is half of the length of the side of the square where O and O_1 are centers of the circles. Find the area of shaded region.

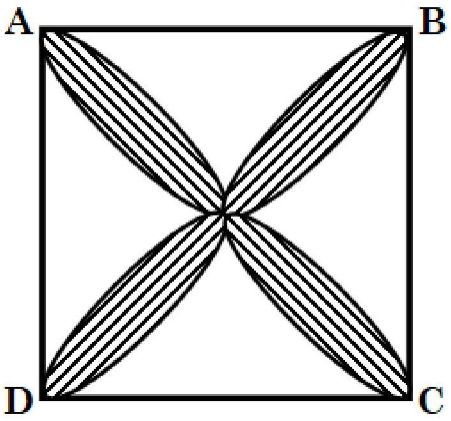


$[3 \times 4 = 12$

SECTION D

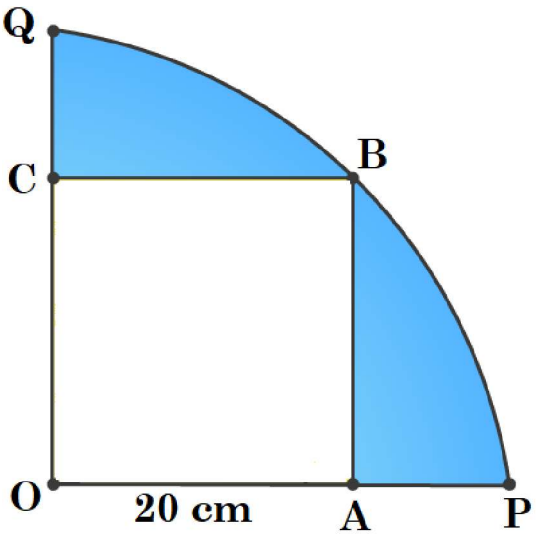
*Followings are of **5 Marks** each (Q17-18).*

Q17. (a) Find the area of the shaded design as shown in the figure, where ABCD is a square of side 12 cm and semi circles are drawn with each side of square as diameter. Use $\pi = 3.14$.



OR

(b) In the given figure, a square OABC has been inscribed in the quadrant OPBQ. If $OA = 20$ cm, then find the area of the shaded region. Use $\pi = 3.14$.



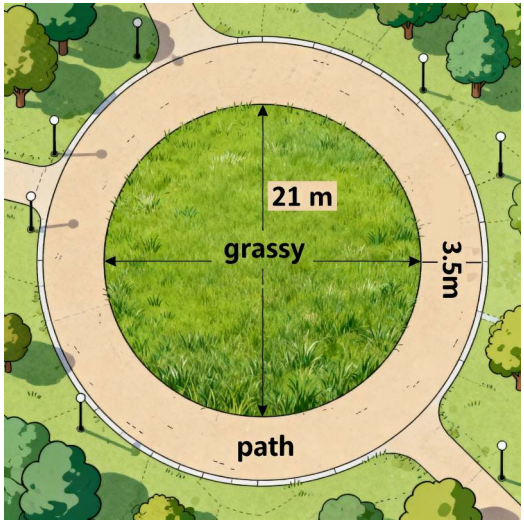
Q18. An athletic track, 14 m wide, consists of two straight sections 120 m long joining semicircular ends whose inner radius is 35 m. Calculate the area of track.

$[5 \times 2 = 10$

SECTION E

*Following is a case-study based question of **4 Marks** (Q19); having three sub-parts (i), (ii) and (iii).*

Q19. **CASE STUDY BASED QUESTION :** A circular park has a radius of 21 m. A concrete path of width 3.5 m is built around the park along its boundary. Many children from nearby localities play inside the grassy part of the park (without the path).



Based on the above information, answer the following questions using $\pi = \frac{22}{7}$.

(i) Find the area of the grassy part (without the path).

(ii) Find the area of the concrete path.

(iii) (a) If the path is to be covered with tiles costing ₹120 per m^2, find the total cost.

OR

(b) If a circular fountain of radius 3.5 m is built at the centre of the grassy part, then find the remaining area of grass.

$[1+1+2=4$